WEEK 2: BIG DATA, PORTER'S VALUE CHAIN, AND SOCIO-TECHNICAL SYSTEMS

Dr. Davood Wadi

Structured and unstructured data are two primary categories of data that differ in their **organization**, **format**, and **ease of analysis**.

Structured Data

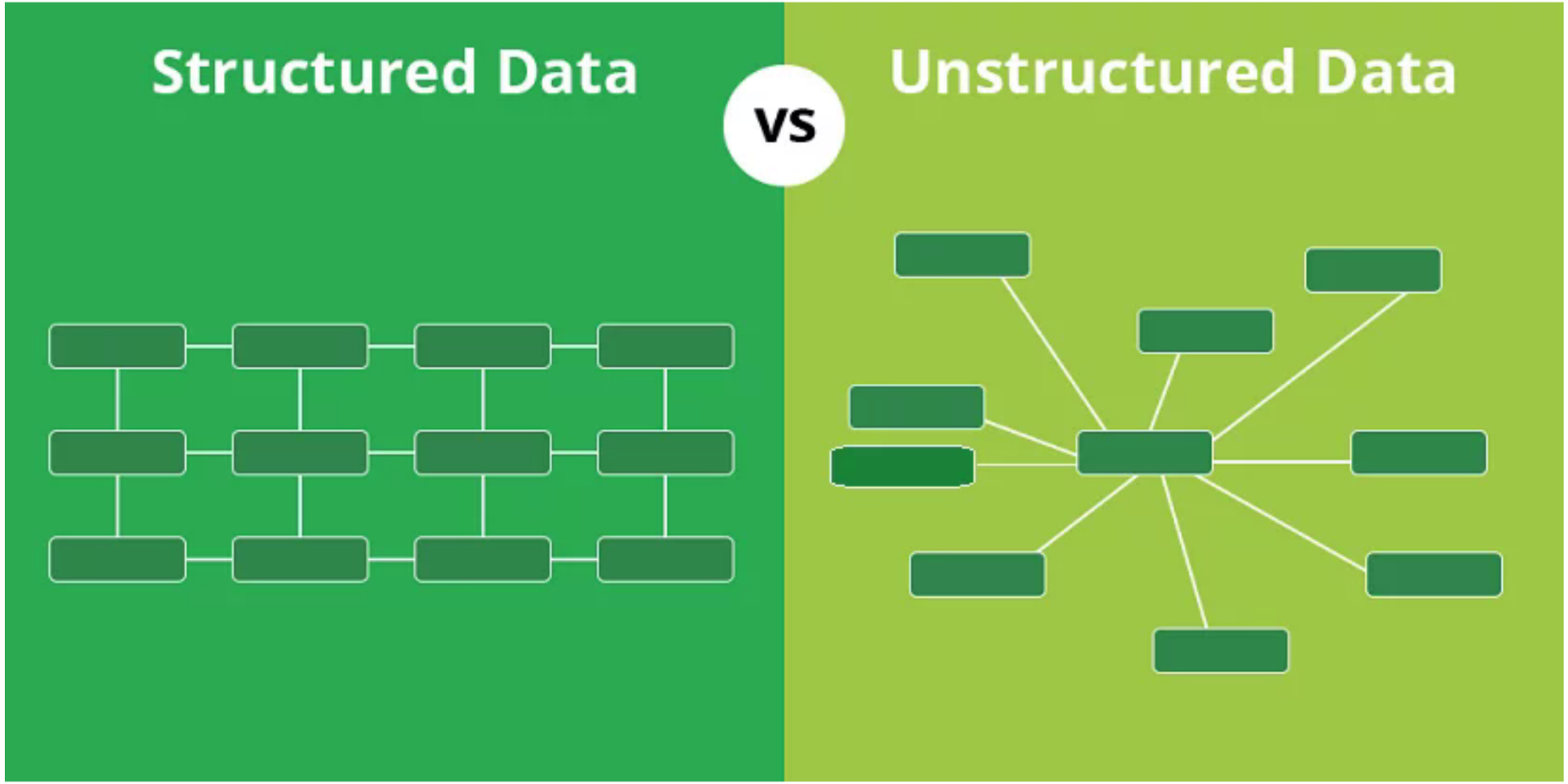
Structured data refers to information that is organized in a predefined manner, typically in **rows** and **columns**, making it easily *searchable* and *analyzable*.

Examples: - Customer databases (e.g., names, addresses, phone numbers) - Transaction records (e.g., sales data) - Inventory lists - Sensor data in a structured format

Unstructured Data

Unstructured data refers to information that does **not** have a predefined *format* or *structure*.

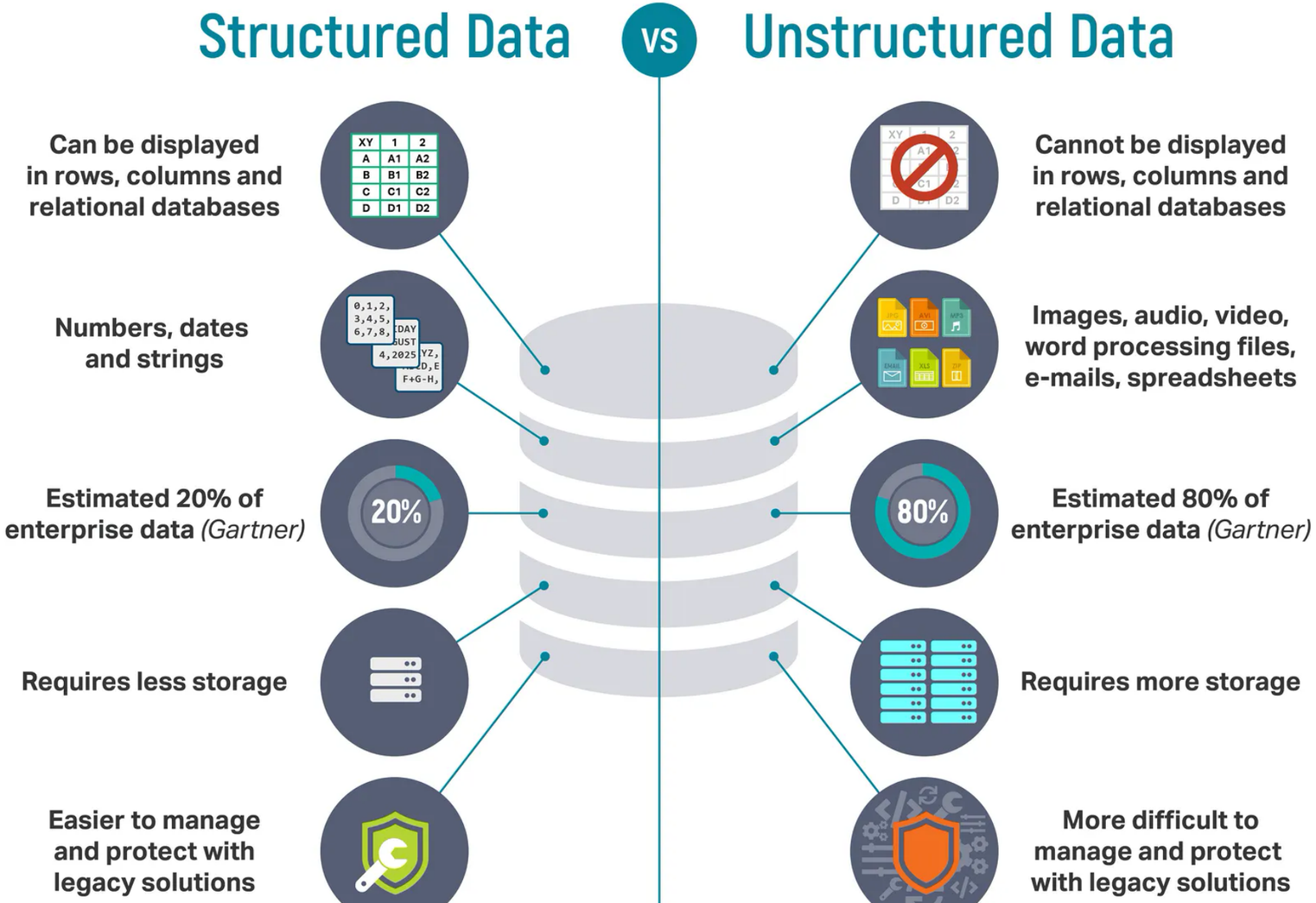
Examples: - Social media posts and comments - Emails and text messages - Images and videos - Web pages and blogs - Sensor data in raw format



Why is it important to distinguish between Structured and Unstructured Data?

How to distinguish between Structured and Unstructured Data

Why did companies focus more on Structured data vs. Unstructured data?



Big Data

Big Data refers to the vast volumes of **structured** and **unstructured** data that are generated at high velocity from various sources, including social media, sensors, devices, transactions, and more.